Non-Gate Quantum Order Finding via Coherence-Driven Closure Selection

Andrei T. Patrascu
FAST Foundation, Destin, Florida 32541, USA
andrei.patrascu.11@alumni.ucl.ac.uk

Abstract

Closure selection is a population dynamics in which defect suppression and compatibility reinforcement select a surviving sector. For modular order finding, its compatibility kernel collapses exactly onto residue classes modulo the unknown order r. We prove an explicit convergence rate and a closed-form basin boundary showing when self-reinforcement instead stabilizes a wrong sector. We then separate three computational regimes. Any realization diagonal in the configuration basis—including dissipative filtering, two-copy postselection, and diagonal measurement feedback—is exactly a classical rejection sampler with success probability of order $1/r$. A coherent defect Hamiltonian plus a mixer produces an exactly two-level continuous-time search. An adaptive measurement–feedback controller, requiring no prior knowledge of r, recovers the order by a gcd readout with expected predicate-level cost $O(\sqrt{r})$; adding a quantumly addressable relational table gives the Brassard–Høyer–Tapp balance $O(r^{1/3})$ with $O(r^{1/3})$ stored values. These are query-model statements: coherent modular exponentiation is still required, and the relational projector requires coherent membership testing. Under dephasing at rate Γ, the model interpolates between the coherent and classical endpoints: the $O(\sqrt{r})$ regime requires $\Gamma/E = O(r^{-1/2})$, whereas fixed Γ/E gives linear scaling with an $O(E/\Gamma)$ gain over classical sampling. Finally, Fourier readout of the one-bit defect predicate succeeds only with probability $(2/r)(1 - 1/r)$, distinguishing this access model from the value oracle used by Shor and by recent low-gate factoring algorithms. State-vector and symmetry-reduced simulations verify the analytic reductions and all reported scaling laws.

Keywords: closure selection; order finding; continuous-time quantum search; dynamic circuits; measurement feedback; dephasing; query complexity

1 Introduction

In the circuit model, order finding is solved by preparing a superposition over exponents, computing $a^x \bmod N$ coherently, and extracting the hidden period r with the quantum Fourier transform (QFT) [1, 2]. Recent work has sharpened this landscape rather than displaced it: exact order finding is possible when a multiple of the order is supplied [3]; Regev's multidimensional factoring algorithm reduces the gate count per quantum run [4]; Ekerå and Gärtner extend

that construction to discrete logarithms and explicitly discuss order finding [5]; and Pilatte has supplied an unconditional correctness proof for this recent family [6]. Those algorithms exploit value-level periodic structure and retain polylogarithmic complexity in the integer size. The purpose here is different: to determine exactly what a population-selection dynamics can obtain from coherence when it uses either a one-bit defect predicate or equality with a stored table of values.

The scheme is closure selection. A population p_t over a finite configuration space X is updated by

$$p_{t+1}(d) \;\propto\; p_t(d)\, e^{-\gamma D(d)}\, h_{p_t}(d), \qquad h_p(d) = \sum_{d' \in X} p(d')\, e^{\beta K(d,d')}, \tag{1}$$

where D is a defect functional and K measures compatibility between configurations. Defective configurations are suppressed, mutually supported configurations are reinforced, and the selected answer is the closure-consistent sector that survives. For order finding, $K(d, d') = [a^d \equiv a^{d'} \bmod N]$ is computable from the input, its level sets are the residue classes modulo r, and the population dynamics collapses exactly onto the masses of those classes.

The term "non-gate" in the title describes the outer computational protocol: continuous Hamiltonian evolution is interrupted by measurement, reset, and classical feed-forward rather than presented as a compiled gate sequence, and no QFT is used. It does *not* assert that a digital implementation avoids gates. In particular, the projector that marks $a^d \equiv 1 \pmod N$ requires coherent modular exponentiation, and a digital realization of the mixer, measurement, and feedback has a gate cost in addition to the query cost analysed here. Continuous- and discrete-time query models can simulate one another with at most polylogarithmic overhead under standard assumptions [7, 8]; thus the presentation changes the mechanism and resource decomposition, not the complexity class.

The main results are fivefold. First, the class dynamics converges exponentially to the defect-free class with an explicit rate whenever defect suppression dominates self-reinforcement; otherwise every pure sector is stable, and we compute the basin boundary in closed form (Section 3). Second, every diagonal realization—including dissipative filtering, two-copy postselection, and diagonal measurement feedback—is exactly a classical rejection sampler whose success probability is of order $1/r$ (Section 4). Third, a coherent defect Hamiltonian and mixer generate an exactly two-dimensional evolution. An adaptive controller with no knowledge of r obtains a defect-free sample in expected evolution time at most $15\sqrt{r}/E$ and uses logarithmically many predicate evaluations, whereas every classical algorithm with the same predicate-level access requires $r^{1-o(1)}$ queries (Sections 5 and 8). Fourth, storing sampled values and searching coherently for compatibility with that table gives $O(r^{1/3})$ cost and $O(r^{1/3})$ memory (Section 6). This is the collision-finding balance of Brassard, Høyer, and Tapp rather than a new exponent; our result identifies the precise coherent-memory assumption under which closure selection inherits it. Fifth, finite dephasing supplies the missing interpolation between the coherent and diagonal limits: fixed Γ/E preserves only a constant-factor gain, while the square-root law requires a coherence time that grows as $\sqrt{r}/E$ (Section 7).

The contribution is therefore an access-model and resource boundary for a proposed selection mechanism, not a cryptanalytic replacement for Shor or Regev. It consists of the diagonal no-go theorem, the explicit feedback bound with unknown marked fraction, the coherent-memory accounting behind the $r^{1/3}$ regime, the finite-dephasing crossover, and a Fourier lemma separating

Table 1: Position of the present results relative to current algorithms. The complexities shown are in the relevant query model and omit the gate cost of realizing the oracle.

method	coherent access	role of this paper
Shor and recent factoring variants	full value $a^d \bmod N$	polylogarithmic order/factoring methods; external benchmark, not reproduced here
predicate closure loop	one bit $[a^d = 1]$	exact two-level reduction and adaptive $O(\sqrt{r})$ upper bound; classical $r^{1-o(1)}$ lower bound
relational closure loop	membership in a stored value set	BHT-type $O(r^{1/3})$ upper bound with $O(r^{1/3})$ quantumly addressable entries
diagonal closure dynamics	no off-diagonal coherence	exact classical simulation and $O(1/r)$ postselection success

a one-bit defect oracle from the value oracle used in period finding. All asymptotic claims are paired with state-vector or symmetry-reduced numerical controls (Section 9).

2 Relation to current quantum-algorithm research

The comparison most important for interpreting this paper is not "gates versus no gates" but *which oracle is coherent and which memory is addressable*. Table 1 states the boundary. Shor-type and recent low-gate factoring algorithms evaluate the full modular-exponentiation value coherently and use its periodic structure [1, 4–6]. The predicate loop below coherently exposes only the bit $[a^d \equiv 1 \pmod N]$. The relational loop exposes equality with a stored value set; it therefore has the same query–memory mechanism as quantum collision finding [9, 10]. Recent NISQ collision bounds explicitly interpolate between full coherent access, bounded coherent depth, and dephasing [11]; recent time–space lower bounds show that the BHT speed-up uses a quantumly accessible table rather than free classical memory [12]. Because the present oracle is periodic rather than a random function, those collision lower bounds are comparators, not lower bounds for the structured instance.

Measurement and classical feed-forward are now experimentally standard components of dynamic circuits: superconducting demonstrations have embedded them in algorithms [13], generated long-range entanglement with up to 101 qubits [14], and extended probabilistic error cancellation to measurement-based operations [15]. These results establish the operational relevance of the controller assumed below, but they do not remove the need for a coherent oracle or provide the rank-one mixer for free. The paper accordingly separates query complexity, coherent memory, and physical implementation rather than combining them into a single "runtime" claim.

Finally, noisy search is known to lose its asymptotic advantage unless the noise decreases with problem size [16]; time-correlated noise can behave differently from memoryless noise [17]. Our dephasing analysis is not a general noise theorem. It is an exact symmetry reduction for the Hamiltonian used here, evaluated for two explicit Lindblad generators. Its purpose is to locate the closure protocol between the diagonal and fully coherent endpoints and to state the scaling condition under which the $O(\sqrt{r})$ claim remains valid.

3 Closure selection for order finding

3.1 The dynamics

Let X be a finite set, $D : X \to \{0,1\}$ a defect predicate and $K : X \times X \to \{0,1\}$ a symmetric compatibility predicate. With $\beta, \gamma > 0$ and $\kappa := e^{\beta} - 1$, the weight $W(d,d') = e^{\beta K(d,d')}$ defines the support field $h_p(d) = \sum_{d'} p(d')W(d,d')$ of a distribution p, and the closure-selection map $\mathcal{F} : \Delta(X) \to \Delta(X)$ is

$$\mathcal{F}(p)(d) = \frac{p(d)\, e^{-\gamma D(d)}\, h_p(d)}{\sum_{x \in X} p(x)\, e^{-\gamma D(x)}\, h_p(x)}. \tag{2}$$

The map preserves the simplex and never increases the support of p. Expanding to first order in a time step gives the replicator equation $\dot{p}(d) = p(d)\,[f_p(d) - \langle f_p \rangle_p]$ with state-dependent fitness $f_p(d) = -\gamma D(d) + \log h_p(d)$ [18]. A fixed point p^* is characterized by $e^{-\gamma D(d)} h_{p^*}(d)$ being constant on the support of p^*: the surviving configurations have equal effective fitness.

3.2 Order finding and sector collapse

Let N be composite, a coprime to N, and r the order of a modulo N. Fix a window $X = \{1, \ldots, M-1\}$ with $M \geq N^2$ and define the defect-free set $F = \{d \in X : a^d \equiv 1 \bmod N\}$, which consists of the $m = \lfloor (M-1)/r \rfloor$ multiples of r in X. The defect is $D(d) = 1 - [d \in F]$ and the compatibility is $K(d,d') = [a^d \equiv a^{d'} \bmod N] = [d \equiv d' \bmod r]$. Both predicates are evaluated by modular exponentiation; neither requires r. The level sets of K are the residue classes $F_\rho = \{d \in X : d \equiv \rho \bmod r\}$, $\rho = 0, \ldots, r-1$, with $F_0 = F$, and we write $P_\rho = \sum_{d \in F_\rho} p(d)$ for the class masses.

Proposition 3.1 (Sector collapse)**.** *For $d \in F_\rho$, $h_p(d) = 1 + \kappa P_\rho$. Consequently the class masses evolve autonomously,*

$$P_\rho^{(t+1)} = \frac{P_\rho^{(t)}\, w_\rho^{(t)}}{\sum_\sigma P_\sigma^{(t)}\, w_\sigma^{(t)}}, \qquad w_\rho^{(t)} = e^{-\gamma D_\rho}\big(1 + \kappa P_\rho^{(t)}\big), \tag{3}$$

with $D_0 = 0$ and $D_\rho = 1$ for $\rho \neq 0$, and the conditional distribution of p_t within each class is invariant in time.

Proof. Split the sum defining $h_p(d)$ into $d' \in F_\rho$, where $W = e^{\beta}$, and $d' \notin F_\rho$, where $W = 1$: $h_p(d) = e^{\beta} P_\rho + (1 - P_\rho)$. Since D and h_p are constant on each class, all members of F_ρ are multiplied by the same factor w_ρ in (2); summing over the class gives (3), and dividing by the class mass shows that the conditional distribution within the class is unchanged. □

The collapse is exact and it reduces an $(M-1)$-dimensional dynamics to r variables. It also fixes the scope of any implementation that tracks class masses explicitly: its state space has dimension r, and r is exponentially large in $\log N$ for the instances that matter in cryptanalysis. In the special regime $r = \mathrm{poly}(\log N)$, r successive modular multiplications already find the order classically. We therefore quantify every cost as a function of r and make no polynomial-in-$\log N$ claim for closure selection.

3.3 Convergence

Theorem 3.2 (Exponential selection of the defect-free class)**.** *Let $P_0^{(0)} > 0$. (i) If $\gamma > \beta$, then for every $\rho \neq 0$ the odds $P_0^{(t)}/P_\rho^{(t)}$ grow at least by the factor $e^{\gamma-\beta}$ per step, so $P_0^{(t)} \to 1$ exponentially from every initial condition. (ii) If the non-target classes carry equal mass initially, they do so at all times, and with $Q_t = P_0^{(t)}$ the odds $\mathcal{R}_t = Q_t/(1-Q_t)$ satisfy $\mathcal{R}_{t+1} = G(Q_t)\mathcal{R}_t$ with*

$$G(Q) = e^{\gamma}\,\frac{1+\kappa Q}{1+\kappa(1-Q)/(r-1)} \;\geq\; e^{\Lambda}, \qquad \Lambda := \gamma - \log\Bigl(1+\frac{\kappa}{r-1}\Bigr). \tag{4}$$

If $\Lambda > 0$ then $\mathcal{R}_t \geq e^{\Lambda t}\mathcal{R}_0$, and $P_0^{(t)} \geq 1-\delta$ as soon as $t \geq \Lambda^{-1}\bigl[\log \mathcal{R}_0^{-1} + \log((1-\delta)/\delta)\bigr]$.

Proof. From (3), $P_0^{(t+1)}/P_\rho^{(t+1)} = e^{\gamma}\,\frac{1+\kappa P_0}{1+\kappa P_\rho}\,P_0^{(t)}/P_\rho^{(t)}$. Since $1+\kappa P_\rho \leq e^{\beta}$ and $1+\kappa P_0 \geq 1$, the factor is at least $e^{\gamma-\beta}$, which gives (i). For (ii), equal non-target masses are preserved because all non-target classes have the same weight, and substituting $P_\rho = (1-Q)/(r-1)$ gives G; the bound $G \geq e^{\Lambda}$ follows from $1+\kappa Q \geq 1$ and $1+\kappa(1-Q)/(r-1) \leq 1+\kappa/(r-1)$. The step count follows by solving $\mathcal{R}_0 e^{\Lambda t} \geq (1-\delta)/\delta$. □

From the uniform distribution on X the class masses are equal up to $O(1/M)$, so (ii) applies with an error that is negligible for $M \geq N^2$; the simulations of Section 9 start from the exactly uniform class vector. If the class masses are only estimated, with errors bounded by ε, the growth exponent is reduced by at most $\kappa(1+1/(r-1))\varepsilon$, since $\log G$ is Lipschitz in Q with that constant; convergence persists as long as the reduced exponent stays positive.

3.4 The competing attractor

Compatibility reinforcement is self-reinforcement: a class is favoured in proportion to its own mass. When $\gamma < \beta$ this creates attractors other than the correct one, and a convergence claim that ignores them is incomplete. The following theorem locates the boundary exactly.

Theorem 3.3 (Basin of the correct sector)**.** *Let $\gamma < \beta$. (i) Every pure sector state $P_\rho = 1$ is asymptotically stable for the class dynamics* (3)*, including the wrong sectors $\rho \neq 0$. (ii) On the invariant two-sector subsystem $P_0 = Q$, $P_\rho = 1-Q$ the fixed point*

$$Q^* = \frac{e^{\beta} - e^{\gamma}}{(e^{\beta}-1)(e^{\gamma}+1)} \tag{5}$$

separates the basins: $Q_t \to 1$ if $Q_0 > Q^*$ and $Q_t \to 0$ if $Q_0 < Q^*$. (iii) On the symmetric subsystem of Theorem 3.2(ii) the boundary is $Q_r^* = \bigl(1+\frac{\kappa}{r-1} - e^{\gamma}\bigr)\big/\bigl(\kappa\,(e^{\gamma}+\frac{1}{r-1})\bigr)$, which lies in $(0,1)$ exactly when $\Lambda < 0$; for $\Lambda > 0$ the correct sector wins from every symmetric start.

Proof. (i) Near $P_\rho = 1$, a small mass P_0 in the correct class changes by the factor $e^{\gamma}(1+\kappa P_0)/(1+\kappa P_\rho) \to e^{\gamma-\beta} < 1$, and a small mass in another wrong class σ changes by $(1+\kappa P_\sigma)/(1+\kappa P_\rho) \to e^{-\beta} < 1$; both perturbations decay geometrically. (ii) The two-sector subsystem is invariant because the update never creates mass in an empty class. Its odds map is $\mathcal{R} \mapsto G_2(Q)\mathcal{R}$ with $G_2(Q) = e^{\gamma}(1+\kappa Q)/(1+\kappa(1-Q))$, which is strictly increasing in Q and equals 1 exactly at Q^* of (5); $Q^* \in (0,1)$ because $\gamma < \beta$. For $Q_0 > Q^*$ we have $G_2 > 1$, so Q_t increases monotonically

and, being bounded, converges to a fixed point above Q^*, which can only be 1; the case $Q_0 < Q^*$ is symmetric. (iii) The same argument applied to G of (4), whose unique zero of $\log G$ is Q_r^*; the condition $Q_r^* > 0$ is $e^\gamma < 1 + \kappa/(r-1)$, that is $\Lambda < 0$. □

Theorem 3.3 is a limitation and a design rule at once. For order finding the defect predicate already singles out the correct class, so compatibility reinforcement adds nothing to the selection and can only create false attractors; the safe regime is $\gamma > \beta$, and the strongest selection is obtained by defect suppression alone.

4 Diagonal realizations are classical

The update (2) is nonlinear in p. Three ways to realize it physically have been proposed: a dissipative channel that removes defective configurations, a selective completely positive map acting on two copies $\rho \otimes \rho$ whose measurement outcome is post-selected, and a single-copy surrogate in which a classical memory of estimated class masses replaces the second copy and drives a feedback bias. All three share a structural property: every operator involved is diagonal in the configuration basis $\{|d\rangle\}$, and every measurement is made in that basis. The following theorem shows that this property makes them classical.

Theorem 4.1 (Diagonal realizations are rejection samplers)**.** *Let a protocol consist of a sequence of quantum operations on $\mathcal{H}_X$ or $\mathcal{H}_X^{\otimes 2}$, each either (a) a trace-non-increasing operation with Kraus operators diagonal in the configuration basis, possibly chosen by classical feedback from earlier outcomes, or (b) a measurement in the configuration basis, possibly post-selected, applied to an initial state that is diagonal in that basis. Then the joint distribution of all measurement outcomes, including the event that every post-selection succeeds, equals that of the classical process which samples the initial configuration(s) from the diagonal of the initial state and, at each operation of type (a), keeps the configuration d with probability $\sum_k |A_k(d)|^2$ (the diagonal weight of the Kraus operators A_k). If, moreover, each step weights every defective configuration by at most $e^{-\gamma}$ and every defect-free one by at most* 1*, then the probability that t steps all succeed is at most $P_0^{(0)} + (1 - P_0^{(0)})\, e^{-\gamma t}$.*

Proof. A diagonal Kraus operator maps $|d\rangle\langle d|$ to $|A_k(d)|^2 |d\rangle\langle d|$, so a diagonal state remains diagonal and its entries are multiplied by the diagonal weights; a configuration-basis measurement of a diagonal state reveals d with probability equal to its entry. Feedback only selects which diagonal operation is applied next, which the classical process reproduces by conditioning on the same outcomes. The two-copy case is the same statement on $X \times X$. For the bound, the unnormalized weight of a defective configuration after t successful steps is at most $e^{-\gamma t}$ times its initial weight, and that of a defect-free configuration at most its initial weight; summing gives the claim. □

Corollary 4.2 (Two-copy selective map)**.** *The two-copy construction with defect operator $\hat{D} = \sum_d e^{-\gamma D(d)} |d\rangle\langle d| \otimes I$ and compatibility operator $\hat{W} = \sum_{d,d'} W(d,d') |d\rangle\langle d| \otimes |d'\rangle\langle d'|$ rescaled by $e^{-\beta}$ so that it is a valid operation, followed by measurement of the first register, realizes the squared-kernel update $p(d) \mapsto p(d) e^{-\gamma D(d)} \sum_{d'} p(d') W(d,d')^2$ exactly on diagonal states, and the probability that t consecutive updates all succeed from the uniform start is at most $1/r + e^{-\gamma t}$.*

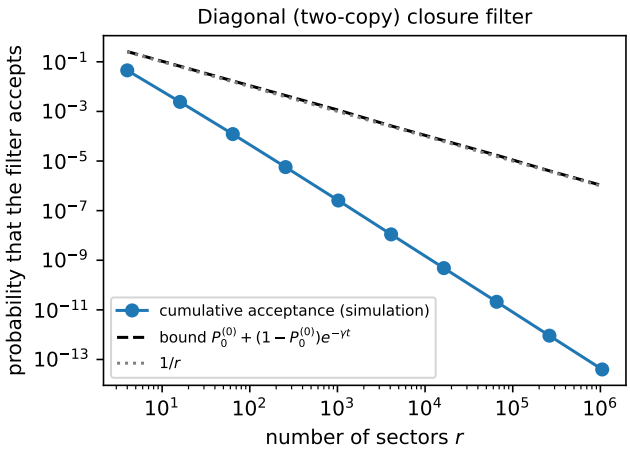


Figure 1: Probability that the trace-non-increasing two-copy closure filter accepts all steps needed to reach $P_0 \geq 0.99$ from the uniform start, as a function of the number of sectors r ($\gamma = \ln 3$, $\beta = \ln 2$), against the bound of Theorem 4.1. The diagonal filter never does better than a single uniform sample.

The squared kernel appears because $\hat{W}$ acts on both sides of the state; the unrescaled operator has norm $e^{\beta} > 1$ and is not a quantum operation. Corollary 4.2 settles the resource question that ensemble accounting alone leaves open: a rejected attempt destroys the state, so the constant acceptance probability of a single step compounds over the $O(\log r)$ steps required for concentration into an overall success probability that is not larger than the probability, $1/r$, of hitting the defect-free set by a single uniform sample. Figure 1 shows the exact cumulative acceptance of the filter for r up to 2^{20}; it lies below the bound and decays faster than $1/r$, because in the squared-kernel dynamics the target class itself carries weight below one while its mass is small. The dissipative channel and the memory-feedback surrogate are diagonal in the same sense: the first removes defective configurations with a rate, the second modifies the weights w_ρ by a classical function of measured class frequencies. Theorem 4.1 applies to both. The conclusion is not that these realizations are useless, since they do implement the selection dynamics faithfully and robustly, in the spirit of dissipative state engineering [19], but that they implement it as a classical stochastic process, and any quantum advantage must come from somewhere else.

5 Coherent closure selection with predicate access

5.1 Where coherence can enter

By Theorem 4.1 the only place where quantum mechanics can change the cost is an operation that is not diagonal in the configuration basis. The defect predicate defines a diagonal operator, the projector $\Pi_F = \sum_{d \in F} |d\rangle\langle d|$, and the natural non-diagonal way to use it is as a Hamiltonian energy penalty combined with a mixing term. We take

$$H = E\big(\Pi_F + |s\rangle\langle s|\big), \qquad |s\rangle = \frac{1}{\sqrt{M-1}} \sum_{d \in X} |d\rangle, \tag{6}$$

which is the continuous-time analogue [20] of Grover's search [21] with the single marked state replaced by the marked set F; E sets the energy scale and $E\tau$ is the continuous-time query cost of an evolution of duration τ. The term $E\Pi_F$ is defect suppression in Hamiltonian rather than dissipative form, and $E|s\rangle\langle s|$ is the mixer. A spatially local driver can also realize continuous-time search, but its performance depends on the graph and dimension [22]; it is not identified here with the rank-one mixer without further analysis. Evaluating Π_F coherently requires reversible modular exponentiation, as in other oracle-based quantum algorithms. Thus "non-gate" refers only to the algorithmic organization—continuous evolution plus feedback and no Fourier transform—and not to a free physical implementation.

Two access models will matter. In predicate-level access the controller learns, for each measured configuration d, only the bit $D(d)$, and the coherent stage uses only Π_F. In value-level access the controller stores the value $a^d \bmod N$ of each measured configuration and can therefore test compatibility of a new configuration with everything it has seen. Section 6 treats the second model.

5.2 Resource ledger

Three resources must be kept separate. First, the theorems below count *oracle time*: $E\tau$ for the coherent marking evolution and one classical predicate evaluation after each measurement. Second, a digital realization has a *logical gate cost*. It must synthesize controlled modular exponentiation for the marking term, implement or simulate the nonlocal projector $|s\rangle\langle s|$, and pay the simulation overhead that converts continuous oracle time into discrete queries and gates [7, 8]. Third, the relational loop has a *coherent-memory cost*: its predicate must test whether $a^d \bmod N$ belongs to a table whose size grows as $r^{1/3}$. This may be implemented by quantum random-access memory or by a reversible lookup network hard-coded from the current classical table; either realization has nonconstant width and depth. The $O(r^{1/3})$ theorem counts queries and table entries, not an end-to-end fault-tolerant gate count. No result below treats these implementation resources as free.

5.3 The two-level reduction

Proposition 5.1 (Exact two-level evolution)**.** *Let $|w\rangle$ be the uniform superposition over F and $x = \langle s|w\rangle = \sqrt{m/(M-1)}$. Under* (6) *the state $e^{-iH\tau}|s\rangle$ remains in* $\mathrm{span}\{|w\rangle, |s\rangle\}$ *and the probability that a configuration-basis measurement lands in F is*

$$P_F(\tau) = \sin^2(Ex\tau) + x^2\cos^2(Ex\tau). \tag{7}$$

The conditional distribution of the outcome within F, and within $X \setminus F$, is uniform.

Proof. Let $|w_\perp\rangle$ be the uniform superposition on $X \setminus F$, so that $|s\rangle = x|w\rangle + y|w_\perp\rangle$ with $y = \sqrt{1-x^2}$. On $\mathrm{span}\{|w\rangle, |w_\perp\rangle\}$ the Hamiltonian is E times the matrix with rows $(1+x^2,\ xy)$ and $(xy,\ y^2)$; its eigenvalues are $1 \pm x$ with eigenvectors $|+\rangle \propto y|w\rangle + (1-x)|w_\perp\rangle$ and $|-\rangle \propto y|w\rangle - (1+x)|w_\perp\rangle$, and $|s\rangle = \sqrt{(1+x)/2}\,|+\rangle - \sqrt{(1-x)/2}\,|-\rangle$. Hence $\langle w|e^{-iH\tau}s\rangle = e^{-iE\tau}\big[x\cos(Ex\tau) - i\sin(Ex\tau)\big]$, whose modulus squared is (7). Uniformity within F and within its complement holds because the state has constant amplitude on each of the two sets. □

Nothing in (6) or in the proof uses r; the two-dimensional structure is a consequence of the predicate alone, and it is checked against a full state-vector evolution in Section 9. Since $x^2 = m/(M-1) \le 1/r$, the marked probability reaches one at $\tau^* = \pi/(2Ex) \approx (\pi/2)\sqrt{r}/E$: the coherent stage concentrates the population on the defect-free sector in a time that grows as $\sqrt{r}$, in contrast to the diagonal filter of Section 4, which reaches the same concentration with probability $1/r$.

5.4 The closure loop with unknown order

The optimal time τ^* depends on r, which is unknown. The closure loop supplies it adaptively. The controller keeps a horizon T, initially $T = 1/E$, and a list of defect-free samples. In each round it draws τ uniformly from $[0,T]$, lets the register evolve under (6) for time τ from $|s\rangle$, measures it, and evaluates the predicate classically on the outcome d. If $D(d) = 0$ the sample is stored and T is kept; otherwise T is multiplied by $\lambda = 6/5$. The loop stops after s defect-free samples and returns $\hat{r} = \gcd(d_1, \dots, d_s)$. The horizon is the controller's memory of the sector support: it encodes the current estimate of the marked fraction, and it is updated only by the measurement record. The choice of a uniformly random τ and geometric growth of the horizon follows the analysis of Grover search with an unknown number of solutions [23], transposed to continuous time. Mid-circuit measurement, reset, and conditional feed-forward have been demonstrated in dynamic superconducting circuits [13, 14], including error-mitigation protocols tailored to measurement-based operations [15]; these experiments support the controller primitives but not the coherent oracle assumed in (6).

Theorem 5.2 (Cost of the predicate-level closure loop)**.** *Let $\lambda = 6/5$ and $r \ge 2$. Until the first defect-free sample, the expected evolution time satisfies*

$$\mathbb{E}[\text{time}] \;\le\; \frac{18}{Ex}, \qquad \mathbb{E}[\text{time}] \;\le\; \frac{15}{Ex} \;\; \text{if } r \ge 4, \qquad \frac{1}{x} \le \sqrt{r}\left(1 - \frac{r}{M-1}\right)^{-1/2}, \tag{8}$$

and the expected number of rounds, each of which costs one predicate evaluation, is at most $\log_\lambda(2/x) + 5$. The expected cost of collecting s samples is $O\big(s(\sqrt{r}/E + \log r)\big)$.

Proof. Since $P_F(\tau) \ge \max\{x^2, (1-x^2)\sin^2(Ex\tau)\}$, averaging (7) over $\tau \sim U[0,T]$ gives, for a round with horizon T, a success probability $p(T) \ge \max\big\{x^2,\, (1-x^2)\big[\tfrac{1}{2} - \sin(2ExT)/(4ExT)\big]\big\} \ge \max\big\{x^2,\, (1-x^2)\big[\tfrac{1}{2} - 1/(4ExT)\big]\big\}$. Let $T^* = 2/(Ex)$. For $T \ge T^*$ the bracket is at least $3/8$, so $p(T) \ge \max\{x^2, \tfrac{3}{8}(1-x^2)\} \ge 3/11 > 1/4$ for every x, and $p(T) \ge 9/32$ when $x^2 \le 1/4$, which holds for $r \ge 4$; write p for the applicable bound. The horizons are $T_j = \lambda^j/E$; let j^* be the first index with $T_{j^*} \ge T^*$, so that $T_{j^*} < \lambda T^*$ and $j^* \le \log_\lambda(2/x) + 1$. The rounds before j^* take expected evolution time at most $\sum_{j<j^*} T_j/2 \le T_{j^*}/(2(\lambda-1)) < \lambda T^*/(2(\lambda-1)) = 6/(Ex)$. The round $j^* + i$ is reached with probability at most $(1-p)^i$ and takes expected time at most $\lambda^{i+1}T^*/2$, so the rounds from j^* on take at most $(\lambda T^*/2)\sum_i [\lambda(1-p)]^i = (\lambda T^*/2)/(1-\lambda(1-p))$, which equals $12/(Ex)$ for $p = 1/4$ and is below $8.8/(Ex)$ for $p = 9/32$, where $\lambda(1-p) = 69/80$. Adding the two contributions gives $18/(Ex)$ and $15/(Ex)$ respectively. The number of rounds is at most j^* plus a geometric number with mean at most $1/p \le 4$, hence at most $\log_\lambda(2/x) + 5$ in expectation. The bound on $1/x$ follows from $m \ge (M-1)/r - 1$. After the first success the horizon never decreases, and the same argument bounds the expected time and the expected

number of rounds for each further sample by constant multiples of the same quantities, which gives the statement for s samples. □

The constants are proof constants; the simulations of Section 9 give about $3.3\sqrt{r}$ for the first sample, predicate evaluations included. With $M \geq N^2$ and $r < N$ the correction factor in Theorem 5.2 is $1 + O(1/N)$.

5.5 Readout

Proposition 5.3 (gcd readout)**.** *Let $d_1, \ldots, d_s$ be independent uniform samples from F, so that $d_i = r k_i$ with k_i uniform on $\{1, \ldots, m\}$. Then $\hat{r} = \gcd(d_1, \ldots, d_s) = r \gcd(k_1, \ldots, k_s)$ is always a multiple of r, and $\Pr[\hat{r} = r] \to \zeta(s)^{-1}$ as $m \to \infty$, where ζ is the Riemann zeta function; for $s = 8$ the limit is 0.996.*

Proof. The factorization is immediate. The probability that s independent uniform integers in $\{1, \ldots, m\}$ are coprime tends to $\zeta(s)^{-1}$ [24]. □

Uniformity of the samples within F is guaranteed by Proposition 5.1 and, for the class dynamics, by the invariance of the conditional distribution in Proposition 3.1. A single defect-free sample already determines r up to the classical factorization of d_1, which is subexponential in $\log N$; the gcd readout with $s = O(1)$ samples is the fully polynomial post-processing and it is what the simulations use.

6 Coherent closure selection with relational memory

In the loop of Section 5 the population memory is a single number, the horizon. Closure selection in its intended form reinforces a configuration according to its compatibility with the whole current population, and the natural memory for that is relational: the controller stores the configurations it has measured together with their values $a^d \bmod N$. Let $\mathcal{M} = \{(d_1, v_1), \ldots, (d_k, v_k)\}$ be such a memory obtained from k uniform samples of X. The support field of the memory population is $h_{\mathcal{M}}(d) = k + \kappa \#\{i : K(d, d_i) = 1\}$, and the set of configurations supported above the baseline is

$$F_{\mathcal{M}} = \{d \in X : \; h_{\mathcal{M}}(d) > k, \; d \notin \{d_i\}\} = \{d : a^d \bmod N \in \{v_i\}\} \setminus \{d_i\}, \tag{9}$$

the union of the residue classes represented in the memory, minus the memory itself. A configuration $d \in F_{\mathcal{M}}$ compatible with the memory entry d_i yields $d - d_i \in r\mathbb{Z}$, a defect-free difference. The relational closure loop runs the coherent stage of Section 5 with Π_F replaced by $\Pi_{F_{\mathcal{M}}}$: coherence amplifies the configurations that the population supports, and the memory grows when amplification fails. Crucially, applying $\Pi_{F_{\mathcal{M}}}$ coherently is stronger than performing a classical lookup after measurement. It requires a reversible membership oracle for the current table—for example, quantumly addressable classical memory or a compiled reversible search tree. The theorem below charges k stored input–value pairs and coherent membership queries; it does not claim unit-cost RAM independent of k.

Concretely, the loop proceeds in levels $\ell = 0, 1, 2, \ldots$. At level ℓ the memory is brought to $k = 2^\ell$ entries by uniform sampling and classical evaluation, a sample whose value is already stored being itself compatible with the memory and therefore recorded as a found difference,

so that stored values are distinct; the horizon is reset to $1/E$, and rounds of the adaptive search with predicate $F_{\mathcal{M}}$ are run, each followed by a classical comparison of the measured value with the memory, until either a compatible configuration is found or the evolution time spent in the level exceeds $4k/E$. Once a level has produced a compatible configuration, further samples are collected at the same level. The gcd of the differences is returned.

Theorem 6.1 (Cost of the relational closure loop)**.** *With $M \geq N^2$, the expected cost until the first defect-free difference is $O(r^{1/3})$; explicitly it is below $160\, r^{1/3}$ up to a factor $1 + O(1/N)$ and $O(\log^2 r)$ classical comparisons, and the expected memory size at termination is $O(r^{1/3})$. Collecting s differences costs $O(s\, r^{1/3})$.*

Proof. Because stored values are distinct, the k memory entries lie in k distinct residue classes, and a sample whose value is already stored ends the level with a defect-free difference, so reaching k entries costs at most k evaluations. Every class contains at least m configurations, hence $|F_{\mathcal{M}}| \geq k(m-1)$ and $x_k^2 = |F_{\mathcal{M}}|/(M-1) \geq (k/r)(1-O(r/M))$. By Theorem 5.2, applied with the predicate $F_{\mathcal{M}}$, the expected evolution time of the search at the level is at most $18/(E x_k) \leq 18\sqrt{r/k}$ up to the stated factor. By Markov's inequality the search exceeds the budget $4k$ with probability at most $18\sqrt{r/k}/(4k)$, which is at most $1/2$ once $k \geq k^* := 9^{2/3} r^{1/3} < 4.33\, r^{1/3}$, and at most $2^{-1-3i/2}$ at the i-th level after that. Let ℓ^* be the first level with $2^{\ell^*} \geq k^*$, so $2^{\ell^*} < 2k^*$. The cost of level ℓ is at most $k + 4k = 5 \cdot 2^{\ell}$ plus $O(\log k)$ classical comparisons, so the levels up to ℓ^* cost at most $5 \cdot 2^{\ell^*+1}$, and the levels beyond cost at most $5 \cdot 2^{\ell^*} \sum_{i \geq 1} 2^i \prod_{i'<i} 2^{-1-3i'/2} < 1.5 \cdot 5 \cdot 2^{\ell^*}$. Altogether the expected cost is below $17.5 \cdot 2^{\ell^*} < 35k^* < 152\, r^{1/3}$, plus $O(\log^2 r)$ comparisons. Each further difference at the final level costs at most the search cost at that level, $O(\sqrt{r/k^*}) = O(r^{1/3})$. □

The mechanism is that of the quantum collision algorithm of Brassard, Høyer, and Tapp [9]: a table of k values and a coherent search for a configuration whose value is in the table, balanced at $k \sim r^{1/3}$. The exponent is therefore inherited, not new. What closure selection contributes is the derivation of the marked set from the compatibility field and an explicit statement of the memory boundary: with only the feedback horizon the exponent is $1/2$, while $1/3$ requires $\Theta(r^{1/3})$ entries addressable inside the coherent predicate. This distinction is consistent with recent collision-finding results that make coherent depth, dephasing, and time–space tradeoffs explicit [11, 12]. Those lower bounds concern generic or label-symmetric collision problems and are not asserted as lower bounds for this periodic oracle.

The classical counterpart of the relational loop is the birthday sampler, which stores values and waits for a repeated one; its cost is $\Theta(\sqrt{r})$, the same as generic classical order-finding algorithms such as baby-step giant-step and Pollard's method [25, 26], and $\Omega(\sqrt{r})$ is a lower bound in the generic-group model [27]. The classical counterpart of the predicate-level loop is memoryless sampling, with cost $\Theta(r)$, and Proposition 8.2 below shows that no classical algorithm with predicate-level access does better than $r^{1-o(1)}$. The two coherent loops gain a square root over their matched classical access models. The relational loop beats the generic classical $\Theta(\sqrt{r})$ scaling by $r^{1/6}$ in query complexity, but its table-lookup implementation cost must be included before making a hardware-level speed claim.

7 Partially coherent regime

The diagonal no-go theorem and the unitary search theorem are endpoints of a one-parameter family. To expose the intermediate regime, let ρ evolve under the Hamiltonian (6) and one of two pure-dephasing generators. Configuration-basis (local) dephasing is

$$\dot{\rho} = -i[H,\rho] - \Gamma\big(\rho - \mathcal{D}_X(\rho)\big), \qquad \mathcal{D}_X(\rho) = \sum_{d\in X} |d\rangle\langle d|\rho|d\rangle\langle d|, \tag{10}$$

whereas sector dephasing removes only coherence between the marked and unmarked subspaces,

$$\dot{\rho} = -i[H,\rho] - \Gamma\big(\Pi_F\rho\Pi_{\bar{F}} + \Pi_{\bar{F}}\rho\Pi_F\big), \qquad \Pi_{\bar{F}} = I - \Pi_F. \tag{11}$$

The rate Γ is the interpolation parameter. Equation (10) destroys both inter-sector and within-sector coherence; Eq. (11) retains coherent uniform modes inside each sector. Neither is claimed to be a universal hardware noise model.

Permutation symmetry within F and $X \setminus F$ closes both master equations exactly on a two-dimensional bright block, augmented by dark populations for local dephasing. For sector dephasing write $p = \rho_{ww}$ and $c = \rho_{ww_\perp}$, where $|w\rangle$ and $|w_\perp\rangle$ are the uniform marked and unmarked states of Proposition 5.1. With $y = \sqrt{1-x^2}$,

$$\begin{aligned} \dot{p} &= -2Exy\ \mathrm{Im}\, c, \\ \dot{c} &= -\big(\Gamma + 2iEx^2\big)c - iExy(1-2p). \end{aligned} \tag{12}$$

The initial condition is $p(0) = x^2$ and $c(0) = xy$. In the overdamped window $Ex \ll \Gamma \ll E$, adiabatic elimination of c gives an incoherent bright-state transfer rate

$$k_\Gamma = \frac{2E^2x^2y^2\Gamma}{\Gamma^2 + 4E^2x^4} = \frac{2E^2x^2}{\Gamma}\,[1 + o(1)]. \tag{13}$$

Thus a constant marked probability requires time $\Theta(\Gamma/(E^2x^2))$, rather than $\Theta(1/(Ex))$ in the coherent limit.

To compare a finite-duration run with sampling, define the known-r cost per marked outcome

$$C^*_\Gamma = \inf_{\tau>0} \frac{E\tau + 1}{P^\Gamma_F(\tau)}, \qquad C_{\mathrm{cl}} = \frac{M-1}{m} = \frac{1}{x^2}, \tag{14}$$

where $P^\Gamma_F(\tau) = \mathrm{Tr}(\Pi_F\rho(\tau))$. Equations (7) and (13) give the scaling law

$$C^*_\Gamma = \begin{cases} \Theta(x^{-1}) = \Theta(\sqrt{r}), & \Gamma \lesssim Ex, \\ \Theta\big((\Gamma/E)x^{-2}\big) = \Theta\big((\Gamma/E)r\big), & Ex \ll \Gamma \ll E, \\ \Theta(x^{-2}) = \Theta(r), & \Gamma \gtrsim E, \end{cases} \tag{15}$$

up to model-dependent constants and the M-window correction. Consequently, a fixed nonzero ratio Γ/E gives linear scaling in r even when it retains an $O(E/\Gamma)$ constant-factor advantage. Maintaining the square-root law requires $\Gamma/E = O(x) = O(r^{-1/2})$, or a coherence time growing

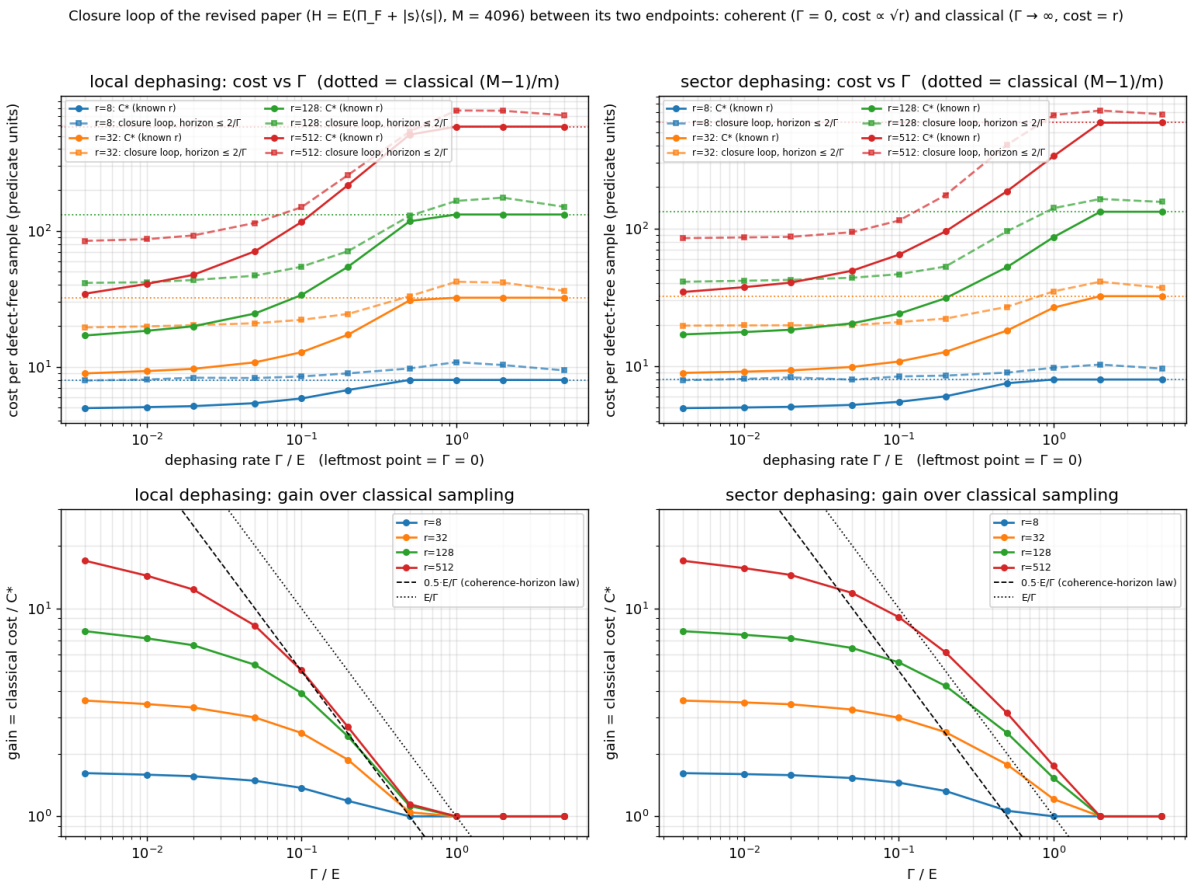


Figure 2: Interpolation between coherent search and classical sampling for $M = 4096$. Solid curves are the optimized known-r cost (14); dashed curves are the adaptive loop with horizon $T \le 2/\Gamma$; dotted horizontal lines are C_{cl}. Local dephasing leaks bright-state population into dark modes and loses the gain earlier than sector dephasing. Bottom panels show that the gain saturates at order E/Γ for fixed dephasing. The leftmost point is $\Gamma = 0$ and is plotted at a small positive abscissa only to display it on the logarithmic axis.

as $\sqrt{r}/E$. This is the closure-model version of the size-dependent noise threshold known for noisy quantum search [16]; it is also consistent with collision-finding results under dephasing [11].

For an unknown order the adaptive horizon must respect the coherence time. We therefore cap it at $T_{\max} = 2/\Gamma$ (and remove the cap when $\Gamma = 0$). Allowing the controller to increase T indefinitely after coherence is lost creates arbitrarily large costs with no corresponding increase in success probability. The cap makes the feedback protocol operationally meaningful and preserves the interpolation: its overhead over C^*_Γ is at most about 2.5 over the tested range.

The two dephasing models separate a physical question that an endpoint-only analysis cannot answer. Preserving within-sector coherence extends the finite-size gain, but it does not change the fixed-Γ/E asymptotic exponent. In the tested range the cost is monotone in Γ: there is no noise-assisted interior optimum. The intermediate regime is therefore real but limited—partial coherence yields a calculable constant-factor advantage, not a new scaling law.

8 What coherence buys, and what it does not

The gains above are polynomial. The following lemma shows why no exponential gain is available with the access that closure selection uses, and locates the source of Shor's exponential advantage.

Lemma 8.1 (Fourier readout of the defect predicate)**.** *Let $M = rL$ and let $g(d) = [r \mid d]$ on $\mathbb{Z}_M$. Prepare $M^{-1/2} \sum_d |d\rangle|g(d)\rangle$, measure the second register and apply the Fourier transform over $\mathbb{Z}_M$ to the first. The probability that the outcome is a nonzero multiple of L, which is the event from which r can be inferred, is $(2/r)(1 - 1/r)$.*

Proof. With probability $1/r$ the second register reads 1 and the first is uniform on the multiples of r, whose Fourier transform is uniform on the r multiples of L; the outcome is nonzero with probability $1 - 1/r$. With probability $1 - 1/r$ the second register reads 0 and the first register is $\big(\sqrt{M}|A\rangle - \sqrt{M/r}|B\rangle\big)/\sqrt{M(1 - 1/r)}$, where $|A\rangle$ is uniform on $\mathbb{Z}_M$ and $|B\rangle$ uniform on the multiples of r. The Fourier transform sends $|A\rangle$ to $|0\rangle$ and $|B\rangle$ to the uniform superposition of the multiples of L, so each of the $r - 1$ nonzero multiples of L has probability $(1/r^2)/(1 - 1/r)$, and their total is $1/r$. Multiplying by the branch probability and adding gives $(1/r)(1 - 1/r) + (1 - 1/r)(1/r)$. □

For a window $M = 2^n$ that is not a multiple of r the peaks broaden in the usual way, and the probability that the outcome lies within $1/(2M)$ of a nonzero fraction j/r is numerically between 0.59 and 0.77 times $2/r$ for $5 \le r \le 101$ (Section 9). The Fourier transform therefore does not help when the oracle returns one bit: a Boolean periodic predicate with a single marked residue per period carries the information of an unstructured search oracle, and a Fourier sample succeeds essentially as often as a uniform sample does. Shor's algorithm uses instead the value register $a^x \bmod N$, whose r distinct values make the function periodic with $\log_2 r$ bits of output structure per query; the QFT converts that value-level structure into a polylogarithmic cost. Recent exact and low-gate variants alter the surrounding algorithm or its resource balance while retaining structured value access [3–6].

The relational loop of Section 6 also uses value-level access, but only through equality with a stored set. Generic collision finding has query complexity $\Theta(r^{1/3})$ [9, 10], and current work shows that the associated speed-up is tied to coherent depth and quantumly accessible memory [11, 12]. The periodic instance studied here is not generic, so those results explain the mechanism and resource tradeoff but do not preclude an algorithm that exploits additional periodic structure. The QFT does precisely that; the relational equality oracle discards it. On the classical side the predicate-level model does admit a direct lower bound, by a divisor-counting adversary.

Proposition 8.2 (Classical predicate-level access)**.** *Let $M \le R^c$ for a constant c. Any classical algorithm that outputs the order from queries to the predicate $g_r(d) = [r \mid d]$, $d \in \{1, \dots, M - 1\}$, and is correct for every $r \le R$, makes at least $R^{1-o(1)}$ queries in the worst case; the same holds for randomized algorithms with success probability at least $2/3$.*

Proof. Let $\tau(d)$ be the number of divisors of d; $\tau(d) \le d^{o(1)}$ [28, Theorem 315], hence $\tau(d) \le R^{o(1)}$ for $d < M$. Consider the run of a deterministic algorithm in which every answer is 0, with query sequence $d_1, \dots, d_q$. Every $r \le R$ that divides none of the d_i produces exactly this run and therefore the same output, so the algorithm is correct for at most one of them; there are at least $R - \sum_i \tau(d_i) \ge R - qR^{o(1)}$ such r. Correctness for all r forces $R - qR^{o(1)} \le 1$, that is $q \ge R^{1-o(1)}$. For a randomized algorithm, fix the random string and take r uniform on $\{1, \dots, R\}$:

Table 2: Order-finding cost as a function of r, separated by access and coherence. These are oracle/query costs; the coherent implementation costs described in Section 5.2 are additional.

access and memory	classical loop	coherent closure loop
predicate bit, any classical memory	$r^{1-o(1)}$, Proposition 8.2	$O(\sqrt{r})$, Theorem 5.2
predicate bit, fixed $\Gamma/E > 0$	$\Theta(r)$	$\Theta((\Gamma/E)r)$ in the intermediate regime
values, relational memory	$\Theta(\sqrt{r})$ (birthday; generic [27])	$O(r^{1/3})$ with $O(r^{1/3})$ coherent table entries
values, Fourier readout	—	poly(log r), Shor and variants [1, 4, 5]
predicate bit, Fourier readout	—	no gain, Lemma 8.1

by the same count the success probability is at most $(qR^{o(1)} + 1)/R$, and by Yao's principle success probability $2/3$ requires $q \geq R^{1-o(1)}$. □

Together with Theorem 5.2, which holds for every r with the same access, this is a quadratic separation up to $r^{o(1)}$ between the coherent closure loop and every classical algorithm in the predicate-level model. Table 2 summarizes the picture. We do not claim a quantum lower bound for structured predicate-level access; the lemma rules out the natural Fourier attack, and the diagonal no-go rules out the selection dynamics itself, as sources of more than a polynomial gain.

9 Numerical verification

All computations are reproduced by the supplementary code, which runs in about ten minutes on a laptop and prints a pass or fail verdict for every control; every number quoted below is read from its output files.

The class dynamics (3) was iterated for $\gamma = \ln 3$, $\beta = \ln 2$ from the uniform class vector. The first two steps for $r = 4$ give $Q_1 = 1/2$ and $Q_2 = 27/34$ exactly. The number of steps to reach $P_0 \geq 0.99$ is 4, 6, 8, 11 and 13 for $r = 4$, 16, 256, 4096 and 65536, within the bounds 8, 8, 10, 12 and 15 of Theorem 3.2, and it grows as $\log r$ (Figure 3a). The basin boundary of Theorem 3.3(ii) was located independently by bisection on the two-class map for $\beta = \ln 6$ and forty values of γ; the closed form (5) agrees with the bisection to machine precision, 2×10^{-16} (Figure 3b).

The two-copy filter of Corollary 4.2 was iterated with rescaled weights while tracking the exact cumulative acceptance of every class. For every r from 2^2 to 2^{20} the acceptance lies below the bound of Theorem 4.1; at $r = 2^{20}$ it is 4×10^{-14} against the bound 10^{-6} (Figure 1).

The coherent loops were tested in two ways. The control, in which r is used nowhere, builds the Hamiltonian (6) on a window of $M = 2048$ configurations from the predicate $a^d \equiv 1 \bmod N$ alone, diagonalizes it, evolves the uniform state, samples measurement outcomes from $|\psi|^2$, and runs the loop of Theorem 5.2 end to end with $s = 8$. For $(N, a) = (15, 2)$, $(21, 2)$, $(91, 2)$, $(221, 3)$ and $(323, 2)$ the recovered orders 4, 6, 12, 48 and 72 equal the brute-force orders, and the marked probability of the full evolution agrees with (7) to better than 5×10^{-14} at twelve evolution times per instance. The scaling runs then use (7) and the uniformity statement of Proposition 5.1 to draw outcomes directly, so that orders up to 65514 can be reached; the controller sees only the outcomes and the predicate. Fifteen semiprime instances with orders from 4 to 65514 were run 24 times each for the predicate-level loop (A), the relational loop (B), memoryless classical sampling (C1) and the classical birthday sampler (C2), with $\lambda = 6/5$ and $s = 8$ throughout. Table 3 lists the mean costs and the fraction of runs in which $\hat{r} = r$; the latter is 0.96 or 1.00 for

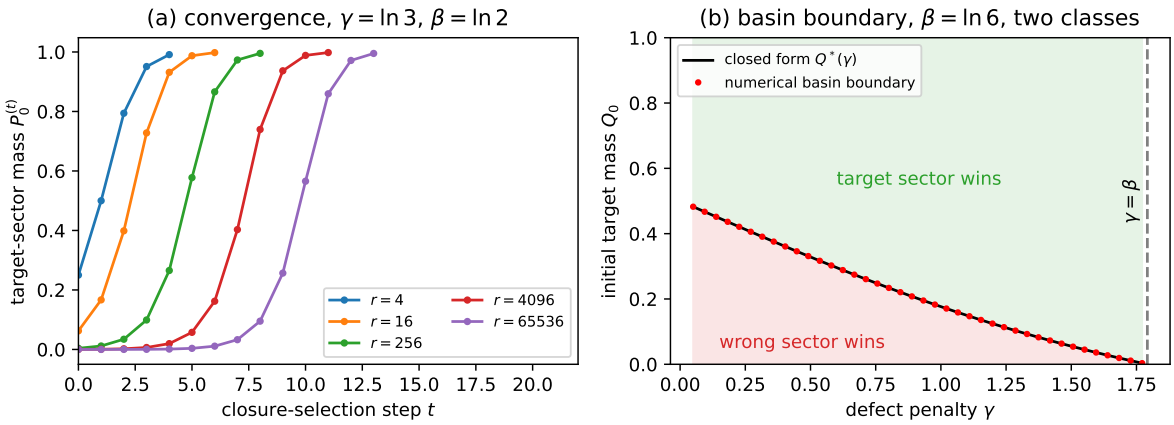


Figure 3: (a) Target-sector mass under the class dynamics from the uniform start for orders $r = 4$ to 65536; the step count grows as $\log r$. (b) Basin boundary of the correct sector for $\beta = \ln 6$ as a function of γ: closed form (5) against bisection on the numerically observed basin. Below the line a dominant wrong sector wins.

every loop and every instance, consistent with $\zeta(8)^{-1} = 0.996$. Power-law fits over $r \geq 64$ give exponents 0.52, 0.32, 0.99 and 0.48 for A, B, C1 and C2, against the predicted $1/2$, $1/3$, 1 and $1/2$ (Figure 4). The prefactors are informative: the coherent predicate-level loop costs about $21\sqrt{r}$ for eight samples and about $3.3\sqrt{r}$ for the first, predicate evaluations included, well inside the proof constant of Theorem 5.2; the relational loop costs about $37\, r^{1/3}$ with a final memory of about $0.75\, r^{1/3}$ entries; and the classical birthday sampler, at about $4.6\sqrt{r}$, is cheaper than both coherent loops for every order in the table, the crossover with the relational loop lying near $r \approx 3 \times 10^5$ in these units. The polynomial advantage of the relational loop is real but asymptotic.

The dephasing computation integrates the exact permutation-reduced master equations for $r \in \{8, 32, 128, 512\}$ and $M = 4096$. Three independent controls pass: the reduced equations agree with direct density-matrix evolution on tractable instances to 10^{-15}; the $\Gamma = 0$ curves agree with Eq. (7) to 10^{-11}; and the local-dephasing cost approaches C_{cl} at large Γ. Table 4 gives the most demanding tested order. At fixed $r = 512$, sector dephasing retains a larger finite-size gain because coherence inside each sector survives, while local dephasing is already indistinguishable from classical sampling by $\Gamma/E = 1$. Across the four orders, the crossover from square-root to linear scaling occurs near $\Gamma/E \sim r^{-1/2}$, as predicted by Eq. (15).

Finally, the Fourier lemma was checked by direct computation of the Fourier-transformed post-measurement states: for $M = rL$ the probability of a nonzero multiple of L equals $(2/r)(1 - 1/r)$ to 10^{-10} for $r \in \{3, 4, 7, 16, 31, 64, 128\}$, and for $M = 2^n$ windows the probability of an outcome within $1/(2M)$ of a nonzero j/r is 0.59, 0.64, 0.69, 0.70, 0.74, 0.75, 0.76 and 0.76 times $2/r$ for $r = 5, 7, 11, 13, 23, 37, 61, 101$.

10 Discussion and limitations

The results delimit the framework, and that delimitation is the contribution. Closure selection over the order-finding kernel is an exact r-sector population dynamics with a computable convergence

Table 3: Mean cost to recover r with $s = 8$ samples, 24 runs per instance, for the coherent predicate-level loop (A), the coherent relational loop (B), classical memoryless sampling (C1) and the classical birthday sampler (C2); the last two columns give the fraction of runs with $\hat{r} = r$ for A and B; for C1 and C2 it lies between 0.96 and 1.00 in every instance. The last row gives power-law fits over $r \geq 64$.

N	a	r	A	B	C1	C2	succ. A	succ. B
579	274	4	37	41	32	12	1.00	1.00
697	38	8	73	48	63	14	0.96	1.00
85	62	16	114	70	130	19	1.00	1.00
6001	4530	32	121	96	248	26	1.00	0.96
1351	253	64	171	120	538	33	1.00	1.00
9509	1481	128	243	168	974	49	1.00	1.00
29041	15011	256	348	221	2141	70	1.00	1.00
20819	18552	513	557	332	3899	96	1.00	1.00
12533	12354	1025	690	347	7828	125	1.00	1.00
49583	40678	2047	1503	385	13105	187	1.00	1.00
49649	222	4095	1586	781	34308	262	1.00	1.00
16637	10238	8190	2763	570	59902	348	1.00	1.00
66013	47415	16375	2626	908	144077	505	1.00	1.00
66013	2301	32750	4226	879	244689	736	1.00	1.00
131753	118594	65514	6086	1265	488898	946	1.00	1.00
fit			$21.1\, r^{0.52}$	$36.9\, r^{0.32}$	$8.1\, r^{0.99}$	$4.6\, r^{0.48}$		

Table 4: Optimized cost C_Γ^* per marked sample at $r = 512$ and $M = 4096$; the classical uniform-sampling cost is $C_{\mathrm{cl}} = 585$.

Γ/E	0	0.05	0.10	0.20	0.50	1.00
local dephasing	34.5	70.8	116	218	511	585
sector dephasing	34.5	49.4	64.5	95.1	186.5	334.9

rate and a computable failure mode. Its diagonal realizations are classical stochastic processes. Its coherent predicate realization is continuous-time amplitude amplification with an unknown marked fraction; its relational realization is a BHT-type collision search whose improved exponent is purchased with coherent access to a growing table. The dephasing family connects these endpoints and shows that a finite coherence advantage need not imply an asymptotic quantum speed-up.

This positioning prevents three possible misreadings. First, the paper does not propose a competitor to Shor's algorithm or to Regev's recent low-gate factoring algorithm. Their structured value oracle and Fourier/lattice post-processing achieve complexity polynomial in $\log N$ [1, 4–6]; the best closure loop here costs $O(r^{1/3})$, which remains exponential in $\log N$ for cryptographic orders. Second, "non-gate" does not mean implementation-free: a physical or digital realization must still pay for coherent modular exponentiation, the mixer, Hamiltonian simulation, reset, measurement, and feedback. Third, the $r^{1/3}$ exponent is not novel as a collision bound; the new statement is which closure memory realizes that known balance and which weaker memory does not.

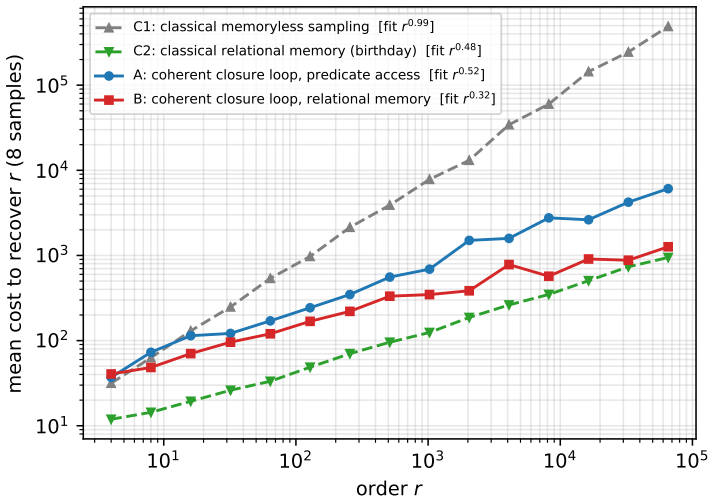


Figure 4: Mean cost to recover the order as a function of r for the four loops of Table 3, with fitted exponents.

Four open limitations remain. (i) We provide no fault-tolerant gate count for either projector. In particular, the rank-one mixer is nonlocal, and replacing it by a local driver is a separate Hamiltonian-design problem. (ii) The coherent table in the relational loop can dominate end-to-end resources; its reversible lookup circuit or QRAM realization must be specified before claiming a practical advantage. (iii) The quantum lower bounds in Section 8 concern unstructured search [29, 30] or generic collision finding. We do not prove that a different algorithm using the structured one-bit periodic predicate cannot beat $\sqrt{r}$; we prove only the classical predicate lower bound, the diagonal no-go theorem, and the failure of the direct Fourier readout. (iv) The dephasing equations model two controlled channels, not correlated control errors, relaxation, leakage, or noisy modular exponentiation. The finite-dephasing scaling is therefore a falsifiable model prediction, not a hardware performance guarantee.

Finally, compatibility reinforcement is redundant for order finding when the unary defect already identifies the answer. Theorem 3.3 shows that excessive reinforcement can even stabilize a wrong sector. Problems with no unary defect may make relational reinforcement essential, but that possibility is not evidence for an advantage until an access model, a non-vacuous positive control, and a separating prediction are supplied.

11 Conclusion

We have reduced closure-selection order finding to a hierarchy of exact or explicitly model-dependent statements. The classical population map converges at rate $\Lambda = \gamma - \log(1+\kappa/(r-1))$ in its safe regime and develops competing attractors outside it. Every diagonal quantum realization is a classical rejection sampler. A coherent defect Hamiltonian and mixer give an exactly two-level search costing $O(\sqrt{r})$ with predicate access; a quantumly addressable relational table gives the known collision-finding balance $O(r^{1/3})$. Under dephasing, fixed Γ/E restores linear scaling, and the square-root law survives only when $\Gamma/E = O(r^{-1/2})$. Fourier readout of the one-bit predicate

has only $O(1/r)$ success, explaining why the result does not inherit the exponential advantage of value-level period finding. The outcome is not a new factoring algorithm but a resource-resolved characterization of what coherence, feedback, and relational memory contribute to closure selection.

Supplementary information

The supplementary package contains the scripts and machine-readable outputs for every figure, table, and numerical control. Deterministic controls are reproduced to the tolerances stated in Section 9; stochastic protocol means are generated from recorded seeds and should be interpreted with their finite-sample Monte Carlo variation.

Statements and Declarations

The author has no competing interests to declare. No external funding was received for this work. All data and code supporting the results are contained in the supplementary package. The author is solely responsible for the conception, analysis, computations, and writing. Generative-AI tools (Anthropic Claude and OpenAI Codex) were used for language editing, code assistance, and independent checks of derivations; every claim and numerical result was reviewed by the author, who takes full responsibility for the content.

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